CHAPTER 5

Conclusion and Abstract

CSP650

Project



suhailayani@uitm.edu.my

What is Conclusion

- Findings, lesson learned from the project
- Limitations
- Recommendations for future work



Conclusion outline

5.1 Conclusion (mapped with all 3 objectives)
5.2 Strength of the systems / Advantages of systems
5.3 Limitation of the systems/ Disadvantages
5.4 Future works/ Recommendations for enhancement
5.5 Summary



SOME GOOD EXAMPLE



Findings, lesson learned (Sample 1)

There are various type of cloud computing services available in the web. This research are separated into 4 distinct objectives. The first objective is to review cloud computing technology for managing artifacts. In reviewing the cloud computing technologies available, a literature review was conducted. Each different cloud services are compared to get the most suitable cloud storage implementation that met the project criteria.

The second objective is to analyze the requirements for developing a system to manage deliverables. In getting the requirement, the stakeholders are given a questionnaire to get their understanding on this subject. Based on the result from the questionnaire, the system requirement are obtained and analyzed. Use cases and domain classes were identified in this phase. All of the result were recorded in a Software Requirements Specification (SRS) report (refer appendix A).

The third objective in this study is to design the system using cloud computing technology. The data gathered from literature reviewing the cloud services and SRS were in the design phase. A thorough design of the system implementation was developed in this phase. This phase also where the system hardware and software requirement are determined. All of the designs and implementation blueprint were compiled in a System Design Documentation (SDD) refer appendix B.



Findings, lesson learned

The final objective is to develop the system in managing deliverables. This part covers the system development and implementation process. It is the most time consuming section of this research. The outcome of this part is a working system that enable the users to manage and collaborate using cloud technology.

At the end of the project, a Post-Usage Evaluation Form was given to a 2 groups of students from ITS670, to give their opinion in using the system. The result from the feedback shows that the system user interface design is easy to understand and simple. In relation with the problem domain, the feedback results shows that the students believe that the system could solve the problem of inconsistency of artifacts files through the practice of cloud storage management, and Collabtifacts provide appropriate tool to edit their project artifacts. The first group have not particular problem with the system and are quite satisfied. However, the other group are giving their concern on their data which are being stored by the system. This might be because the system is relatively new to the students, and they have yet to be associated with it. Thus, this show that the problem of inconsistencies between data could be solve by the system, but we can expect that some students are still reluctant to use a newly developed software for them to trust their project files in it.



Further Work

i. Implementation of UML Diagram collaborative tools, this feature is too big of a scope to be done within the timeline. It could be done as a standalone tools unrelated to Google Drive, while losing the ability to store files at the cloud service.

ii. Page to display list Students to Instructor, list of Instructors to Administrative Instructor.

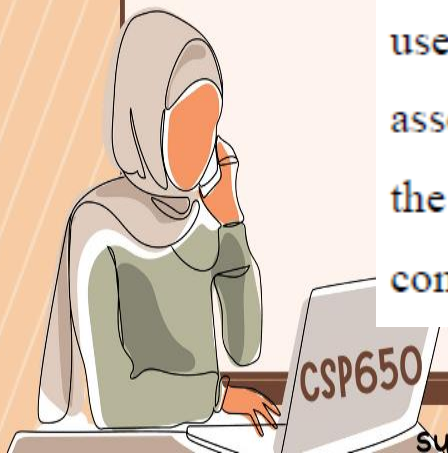
iii. Display of artifact list for students. While the instructor side of artifact lists were able to display at almost all time. The student side of the list varies. At certain time during the execution, the artifact list for students were able to be shown. But at some other time, it does not. The cause for this was not yet determined by the researcher. It could be because of the type of API key used in the system is incorrect.

iv. Deletion process of all objects. This process requires validation of SQL integrity constraints prior to removal of any object that was instantiated. This cannot be done in the time given, as an actual implementation need functional testing to be performed to validate each query. Beside Collabtifacts own database to be validated, the system also need to check whether the Google Drive data is also removed or affected by other constrain by Google. This is a process that was not estimated by the researcher prior to design process.



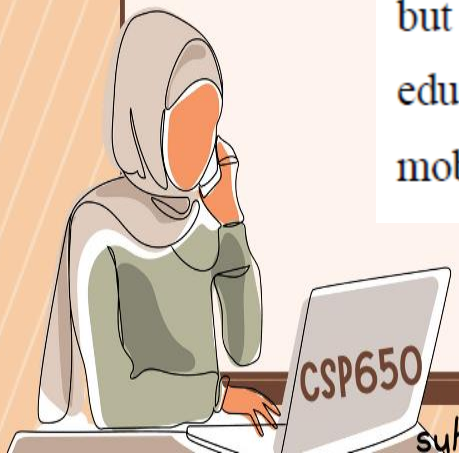
Findings, lesson learned (Sample 2)

As for the evaluation, the evaluation takes place at DAM Johor Bahru Centre. There are two types of evaluation that were conducted which are usability testing and also learning assessment. In this project, think aloud method was used in usability testing for dyslexic student as well as usability questionnaire provided for teachers. For learning assessment, the pre-test and post-test assessments were conducted. Based on the usability testing, the participants disclosed that they find the courseware is easy to use, easy to learn, very useful and they satisfied with it. On top of that, based on the learning assessment, the proposed method shows the increment of marks achieved by the dyslexic student. Thus, it verified that the proposed method is better than conventional learning method.



Further Work

This is because the understanding and comprehension in using the new software is very difficult and challenging. Third, the developers also recommended to prepare more sample devices as it could save more time compared to one sample device only if there are problems occurred during the evaluation phase. Other than that, the developers are recommended to develop the similar courseware with different subject as requested by the teacher. On top of that, as for the current courseware that was developed, the teacher suggest that questions in the courseware should be added more as it proven to be one of the effective learning material for dyslexic student. Last but not least, the developers also recommended to develop similar educational application for other operating system as well as the developed mobile application are only restricted for Android platform only.



Findings, lesson learned (Sample 3)

From the research that had been done, this research learned that human behavior can be categorized under three main aspect of Dooyeweerd study which explained each one of the human behavior. The first aspect covered on human individual aspect, next is social aspect and followed by societal aspect of the human. Through these main aspects of Dooyeweerd, each category of the human behavior was able to be described in more details and clarified. Hence, through Dooyeweerd aspects, each of the causes is to understood and expressed.

Distraction eventually would happen when one lost their focus and their commitment, this can be prove through the uses of pistic aspect of Dooyeweerd. Other than, what made people become distracted while doing their work also because of their health condition, this has come to understanding through the study of physical aspect of Dooyeweerd.

Through the perspective of lingual and social aspect of Dooyeweerd understanding of human nature that lack of communication can be understood. This can be expressed, through these two aspects, it can be seen that the people often make mistakes like accidental deletion or modification when there exist no proper communication and interaction between them, whether among staff or with the upper management.



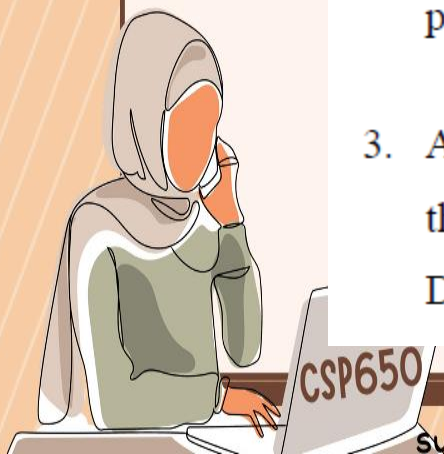
Further Work

The researcher hopes that this study will be useful and helpful to other analyst who wants to analyze on data loss in general and specifically that was made by human. As for the future work, there are some recommendations from researcher which are:

1. The researcher recommends that the future researcher will continue this study by using an ethnography-based research as a research strategy. This can helps researcher to obtained more accurate data and give a depth understanding on research topic.

2. The researcher recommends that the future work of this study to thoroughly study management of the logbook in the organization and could study the pattern of the data loss and errors made by people within the organization.

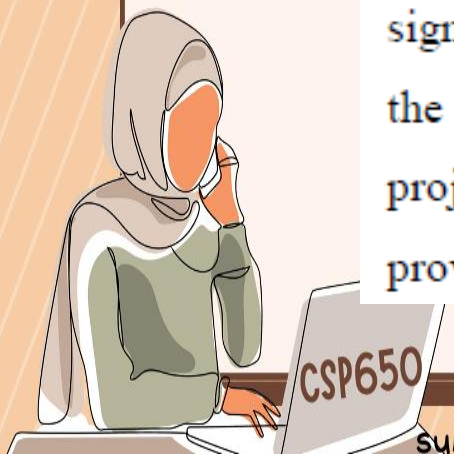
3. Apply the understanding and improve the working environment and evaluate the effectiveness of the study on data loss due to human error using Dooyeweerd aspects in data security in the above mentioned organization.



Findings, lesson learned (Sample 4)

Test case design is a vital task of software testing. However, finding a suitable system to be tested is a pre-requisite to design test case and it has to be done as early as possible. The researchers has learned that taking too long time to find a suitable system will deteriorate the timeline to complete the research project. To ensure that OATS is a systematic approach in fault-finding process, the researcher has strictly followed all of the OATS steps that has been studied earlier. Different steps of OATS depend on the number factors or level that has been identified which has been discussed in previous chapter.

Despite that, it has been proven that design of test cases using OATS is significant to prevent a software tester to execute test cases endlessly while the test coverage remains optimum. The finding and analysis of this research project are represented in well-formed explanation, tables and figures to provide an in-depth understanding for future researcher.



Further Work

1. It is highly recommended to have a good understanding on Orthogonal Array (OA) as it is not originated from software testing domain but it falls under advanced Mathematics theory.
2. It is advisable to explore more on how to identify a suitable system to be tested.
3. The future researcher should study more on OA that dealing with mixed-number of array.
4. The future researcher should identify the steps or activities to create test case manually if it happens to be unable to find available tool for generating test cases.
5. The future researcher may consider to prove that mutation testing is useful to collaborate with OATS.



Sample 5 : Conclusion



suhailayani@uitm.edu

CHAPTER 5

CONCLUSION AND RECOMMENDATION

This chapter elaborates and summarized this research to achieve the objective of the project. The limitations and recommendations are elaborates and describes according to the development of this project. The limitations and recommendations that are described can be used for reference purpose as well as future enhancement.

5.1 Conclusion

Based on the studies and researches conducted in this project, the project is basically about the prediction of the products bought by customers to know which class does the products belongs to. The prediction is based on the customers' preferences. The data mining technique which is Naïve Bayes technique is used to display the probability class of the products as the outcome of the prediction system.

The prediction system will display and suggest for the stakeholder to proceed with promotion of the related products to clear the products in the store for that particular month. The clearance of products in store is done to increase sales in AAIRA. The list of products suggested to be combined is based on the criteria included for the predicted products. If the products entered is predicted as a low sales class, the list of products suggested to be combined is from a high sales class. On the other hand, if the products entered is predicted as a high sales class, the list of products suggested to be combined is from a low sales class. As for the products that is predicted as medium sales class, the list of products suggested to be combined is products from the same medium sales class. However, the combination of the products are based on the criteria of whether purchasing on Weekday, Weekend and Payday.

5.2 Strengths

Every steps and activities conducted throughout the development of this prediction system enhance the strengths of the project. First, the prediction system can display the percentage probability of each class. It can assist the stakeholder to identify which product need to be focused to increase the sales based on Naïve Bayes technique. It also display the suggestion in which what stakeholder could do after identifying the class of the products especially for the low sales products.

The prediction system also provide promotion section that can assist the stakeholder in determining the product that can be combined to increase the sales and clearance of abundance products in store. This would help the stakeholder in managing their inventory more effective and efficiently.

In addition, the prediction system also provide more reliable data storage for the stakeholder to store the information about the products. This can aid the stakeholder to store the required data efficiently in the data storage. The tedious manual process of storing information could be minimized with the system.

Lastly, as the process of developing this prediction system includes the Naïve Bayes technique, it can help and aid the stakeholder, researcher and academician in providing more knowledge and information about the

Sample 5 : Strengths of the system



Sample 5 : Limitation of the system



5.3 Limitations

Every system developed consists of several limitations that prevents from accomplishing fully optimum result of the system. The limitations can be used to improve the findings and results from this project in the future.

The first limitation of the prediction system is that this prediction system only focused on the criteria of products that are being purchased by customers in 2016 and 2017. Since there is possible new products entered by stakeholder into the store in the current year, it must be included into consideration. The prediction system also can only display and produce a prediction of the products based on certain criteria or attributes that are only included in this prediction system.

Next, the scope of the prediction system is only focused for analyzing data of the online seller, AAIRA. Hence, the result and outcome of the prediction system may be biased and only represents the online seller, AAIRA, not all of the online seller who sells fashion clothing. This limitation might affect the accuracy of the prediction result as the prediction system only focused on the current environment of the stakeholder.

Lastly, the intended user for this project is only online seller, AAIRA. The customer who bought and purchased products cannot use the prediction system. Therefore, the process of ordering products by customer are not included in the system.

Sample 5 : Future work of the research

5.4 Recommendation

From the limitations described above, recommendations to solve the limitations can be proposed for future enhancement. The recommendations can be proposed to make the prediction system more reliable.

This prediction system can be enhanced and improved by providing sections to add other criteria or factor that can be used to determine the class of the entered products. By improving and providing functions to add and modify the criteria, this prediction can be useful and assist the user of the prediction system regardless of the changing environment.

Next, the scope of the prediction system can be widen which not only data for fashion clothing can be used for the project. Other data such as fashion shoes and female accessories can also be used to determine the class of the products.

Lastly, the ordering product functions may be added in the system. By having the ordering product functions, the system can be upgraded into e-commerce system that involve both customer and online seller.



Sample 5 : Summary

5.5 Summary

In a nutshell, this chapter concludes and remarks as the last chapter of this project. This chapter summarizes and concludes all activities and objectives that need to be achieved in completion of the project. The limitations that can be discovered during the development of the prediction system is identified which remarks as the weaknesses that need to be fixed and improved in future enhancement of the prediction system. In addition, in conjunction with the limitations discovered during the development of the prediction system, several recommendations are elaborated and discussed to ensure the prediction system can fully assist and more reliable for better improvements.



Breakdown

Chapter	Weight	Percentage of Num of Pages	Eg (Total pages 50)
Chapter 1 – Introduction	2	12.5%	**~ 5**
Chapter 2 – Literature Review	3	20%	**~ 15**
Chapter 3 – Methodology	4	25%	**~ 27**
Chapter 4 – Results/Findings/Prototype	5	30%	
Chapter 5 – Conclusion and Recommendation	2	12.5%	**~ 3**



suhailayani@uitm.edu.my

What is Abstract

- Summarizes the whole project
  - Project motivation
  - Summarize of Problem
  - Summarize of Objectives
  - Methodology
  - Findings

Carries weight - 1



SOME EXAMPLES



ABSTRACT

AAIRA is a sole-proprietorship online seller that sell varieties of products online. AAIRA uses social media such as Facebook and Instagram as a platform to market and sell their products. The process of ordering products by the customers are mainly by using Whatsapp application. The record of the ordered products and customers' details are recorded manually into a journal. However, AAIRA encountered few problems by the current process in predicting the inventory to be ordered when restock as the process involves human judgments, tedious customer's ordering information and products to look into which is time consuming. In order to reduce and minimize the problems, AAIRA need to use the computerized prediction system that helps in predicting the inventory. The aim of this project is to develop a web-based prediction system to assist the process of predicting inventory when restock. AAIRA can predict the next inventory when restock in shorter time and increase the accuracy of prediction. The data mining technique uses for this prediction system is Naïve Bayes where the highest probability of the inventory is selected and need to be focused when restock. The outcome of this prediction system is to display the list of the predicted inventory with the calculated probability. As a result, the average percentage of highest accuracy is 90 per cent. The highest probability of the inventory means that the inventory need to be increased when restock. The lowest probability of the inventory is suggested to be combined with any highest probability inventory to prevent abundance of inventories in store. For future enhancement of this project, the mobile application feature can be added to ease the process of prediction in short time.

Intro

Problem

Aim/objective

Method

Findings

Conclusion/ recommendation



ABSTRACT

ACIS as known as Academic Contemporary Islamic, is one of the unit under Universiti Teknologi Mara (UiTM) which the students can apply zakat during their studies. For instance, in UiTM Kampus Jasin, Melaka, students need to go to ACIS to apply the zakat. However, during the process of applying the zakat, there are several problems that occur such as missing form, time consuming, lack of information dissemination and difficulties to filter student qualification with zakat requirement. The aim of this project is to develop a web-based system that can assist staff of ACIS in choosing the qualification students for zakat using rule based technique which can achieve all the objectives of this project. The methodology that had been used is waterfall methodology. The technique that had been implemented is rule based technique which is the data accepted is transformed into constraint rules. The rules consist of IF-THEN statements which is to produce an accuracy filtering of students and assists staff to manage the student's application effectively and efficiently. As for the result of this project, the system will display the selected students with their types of asnaf and students that are not selected is in waiting list based on the suggested percentages of criteria that showed the highest or lowest tendency of students to get the zakat. The project also has the percentages that shows the particular students is selected and not selected by using explanation facilities provided in rule based architecture. Hence, the staff still can change the criteria based on changing in rules. In the future, it is recommended that the staff can view the record of students, staff management features, inference engine and real-time technology can be added for better enhancement of this system.

Intro

Aim/objective

Findings

Problem

Method

Conclusion/ recommendation



ABSTRACT

The ultimate Frisbee or commonly known as "ultimate", is a sport that combined the elements of American football, basketball and soccer where it is being played using a sports disc. In ultimate, there are two types of tournament which are normal tournament where players register themselves in a team and hat tournament where players register themselves individually. However, hat tournament organizer (TO) found that the current player selection processes have a few weaknesses such as the processes involved human biasness, consume a lot of time and tedious players' data to look into. To reduce these problems, the hat TO use a computerized system that helps in the player selection processes. The aim of this project is to develop a web-based system that can assist Borneo Headhunters as the hat TO in selecting the players using rule-based technique. TO can select the players systematically in a shorter time whereas the tournament players will get to be selected fairly without the biasness of TO. The outcome of this system is that it displays the list of selected players and the reasoning for why the players are selected. This system also displays the names of the unselected players in the waiting list. TO still can modify the selected players if there are changes in rules. In the future, it is recommended that notification feature, team management feature, inference engine and real-time technology can be added for better enhancements of this system.

Intro

Aim/objective

Findings

Problem

Method

Conclusion/ recommendation



Report Presentation

- Well structured
- Follows standard research reporting procedure/format
- Clarity of language
- Consistency of content
- Logical flow
- Use of figurative language for all materials

Carries weight - 2



Report Presentation

Element	Excellent (8-10)	Good (6-7)	Satisfactory (5)	Poor (1-4)
Report Structure & Format	Well structured	Structured	Poorly structured	Not structured
Sentences, Grammar & Flow	Complete No error Sentences flow	Complete Minor errors	Complete Major errors	Poor grammar structure
Figures, graphs, charts and drawings	Accurate Consistent Good quality	Not accurate Consistent	Not accurate Not consistent	Bad quality



To-Do This Week

1. Updated Chapter 4
2. Draft Chapter 5
3. Draft Abstract
4. Submit no1-no3 to Supervisors based on dateline given in *important dates*.



THANK YOU!



suhailayani@uitm.edu.my